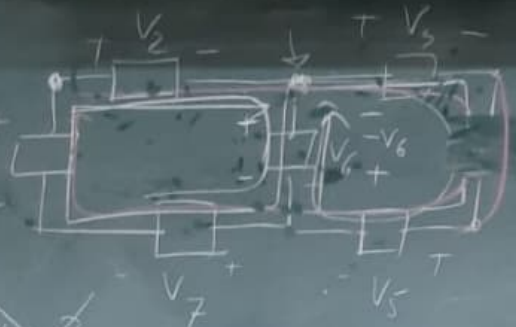


① KVL (Law of Kirchhoff)

$V_3 + V_5 + V_5 - V_6 = 0$

$V_2 + V_6 + V_7 - V_1 = 0$

$V_1 = ?$, $V_6 = ?$

Circuit B elements (branches)

N nodes

KVL #?

eqns $B - N + 1$

② Planar Circuit



Vanishes = B!

a priori $N-1$



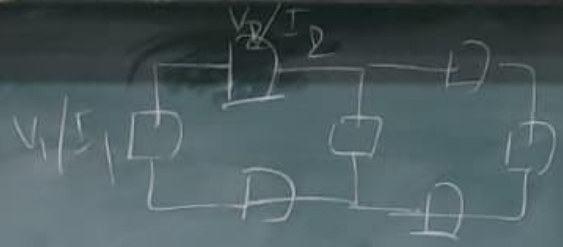
KCL 5 6-1 (Rank of Incidence matrix)

N nodes $N-1$

Vanishes B

a priori $B - N + 1$

PS 3



ETNS for $\underline{V}$ $\Rightarrow$ KVL

$B - N + 1$

$\underline{I}$ $\Rightarrow$ KCL

$N - 1$

Total var. $2B$

All elements Rps

Total eqn B

Eith B eqn

Thanks to Ohm's law

$V_K = I_K R_K$



V_1 I_1 V_2 I_2

$V_1 = 2 I_1$

$I_1 R_1$ $I_2 R_2$

$-3 + V_1 + V_2 + 1 = 0$

$2V + V_3 + 1 = 0$

$I_1 = I_2 + I_3$

$$\begin{cases} -3 + 2 I_1 + I_2 \cdot 1 - 1 = 0 \\ +2V + 2 I_2 - 1 (I_1 - I_2) = 0 \end{cases}$$